

Enhancing Learning Through Virtual Reality and Neurofeedback: A First Step

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ABSTRACT

Virtual reality presents exciting new prospects for the delivery of educational materials to students. By combining this technology with biological sensors, a student in a virtual educational environment can be monitored for physiological markers of engagement or more cognitive states of learning. With this information, the virtual reality environment can be adaptively altered to reflect the student's state, essentially creating a closed-loop feedback system. This paper explores these concepts, and presents preliminary data on a combined EEG-VR working memory experiment as a first step toward a broader implementation of an intelligent adaptive learning system. This first-pass neural time-series and oscillatory data suggest that while an EEG-based neurofeedback system is feasible, more work on removing artifacts and identifying relevant and important features will lead to higher prediction accuracy.

Categories and Subject Descriptors

• **Applied computing~Interactive learning environments** • *Applied computing~Biological networks* • *Applied computing~Computer-assisted instruction* • *Applied computing~Psychology*

Keywords

Virtual reality; EEG; Neurofeedback; Human-computer interaction

1. INTRODUCTION

An important and foundational concept of learning analytics is the necessity of feedback or intervention to improve education and learning. Since the early days of learning theory, such as Kolb's Experiential Learning Theory [1] and Bandura's Social Learning Theory [2], feedback has been considered to be an integral element of learning. A large amount of modern empirical research has focused on the usefulness and appropriateness of different

methods of feedback. In general, feedback seems to be more effective when it focuses the individual's attention on task-related information as opposed to self-related information or praise [3], and when it is immediate as opposed to delayed, particularly when more real-world paradigms are used [4, 5, 6]. Additionally, the most successful feedback provides information on how to improve, not just the correct answer, while remaining relatively simple and easy for the learner to understand [7].

Campbell and Oblinger's five-step model of learning analytics [8] highlights the importance of tailoring interventions for learners in the Predict, Act, and Refine stages. By collecting data about individual learners, teachers can create personalized models that predict success in a particular course, lesson, or even question. If the model reports that a student is at risk for failure, the teacher can intervene to provide additional guidance to the student. Finally, the result of this intervention, and continued collection of data, can be used to refine and update the model for continued successful prediction. These ideas of effectively utilizing data to improve learning interventions are similarly expressed in other learning analytics models as well [9, 10].

Recently, Clow [11] expanded on the five-step model by introducing the Learning Analytics Cycle. Here, learners engage in some form of education material, leading to collection of data (which could be demographic, assessment-based, etc.) about the learner. These data are processed into metrics, outcome variables which provide insight into learning or predict success. The important last step is to use these metrics to intervene and alter the learning process, effectively creating a closed-loop system; the data output by the learner determines the intervention, which affects the input.

Under this framework, learning will be most improved when interventions are provided in real-time to the learner – as soon as some predictive metric is obtained that identifies the individual will fail, the system should intervene. Here, we propose a system to accomplish this goal that can provide rapid, informative intervention prior to the subject even responding. By measuring electroencephalography (EEG) signals, we can track the individual's learning state in real-time. Additionally, by providing instruction in a virtual reality (VR) environment, we can not only provide a more realistic learning environment, but also more easily and adaptively alter the learning environment to intervene based on the EEG signal, effectively closing the loop.

2. VIRTUAL REALITY

VR refers to technology that simulates a realistic three-dimensional environment for the user to interact with. This is typically and most successfully implemented as a headset, or

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head-mounted display, with a screen that the user wears. The digital environment is sent from the computer to the screen in the headset. This allows the user to look around the 3D environment, and also acts to reduce sensory input from the outside world.

Although interest and development of VR technology has been present since the 1980s [12], the technological advancements of the last several years have put a spotlight on VR's potential applications and propelled the industry to new heights [13]. Importantly, many of these advancements, such as improved graphical rendering and room-scale positional tracking, allow for greater "presence" in the environment. Presence, defined as the phenomenological sense of actually being in a virtual environment [14, 15], is important for educational VR applications. Essentially, students will be more focused and attentive to learning material, and less distracted by task-unrelated stimuli, if they are deeply immersed in the virtual environment. This immersion could enhance education in even advanced areas of study – for example, a student studying marine biology could explore virtual ocean environments and interact with simulated marine life.

While the literature is currently somewhat limited, there has been some research on VR applications geared toward training individuals and improving learning [16]. VR has been shown to be useful for learning certain visual-spatial tasks [17], and this knowledge transfers to real world spatial navigation [18, 19]. Some preliminary work has also shown modest benefits on test scores after VR training in geometry [20] and geography [21]. Additional work has shown real world benefits in surgery performance after sessions of virtual surgery training [22, 23]. Lastly, a large-scale meta-analysis of virtual reality studies in educational settings found that VR was overall effective at improving learning outcomes [24]. Thus, while the evidence is still somewhat limited, it is reasonable to believe that VR-based educational systems give benefits to learning.

The additional benefit of a VR educational environment is it allows for a rapid, automatic, and personalized feedback system. Environments can be created that are tailored for each individual based on personality factors, learning styles, and level of knowledge about a subject. Additionally, the environment itself can be altered in real-time based on the learner's current experience, attentional state or level of engagement, performance on the task, or other cognitive or biological shifts that might cause changes in behavior. There are a large number of specific environmental changes that could be used, and which choice would be optimal for improving learning would likely depend on the particular task, as well as the individual. However, based on research on effective feedback, these VR interventions likely should rapidly direct attention to highlight important information about the task at hand.

In order to provide these effective VR interventions, data that is predictive of success must be collected from the learner. Ideally, multiple measures would be extracted in order to improve accuracy, as a single measure is unlikely to be sufficient. However, data based on behavioral output from the learner, such as a response on a test, may not be informative as to the actual internal mental state of the individual. A better tool to measure this is EEG.

3. ELECTROENCEPHALOGRAPHY

EEG is a non-invasive neuroimaging method that measures electrical activity from the brain through sensors or electrodes

placed on the scalp [25]. The measured brain activity represents the summed postsynaptic potentials (both excitatory and inhibitory) of thousands of neurons whose firing is synchronous in time [26, 27]. Given issues of volume conduction and spread of electrical signals, the majority of EEG signals are likely generated by cortical pyramidal neurons, although it remains possible that subcortical structures do contribute [28]. Successful measurement of EEG signals also depends on the spatial orientation of firing cells – neurons that are parallel to each other and perpendicular to the cortical surface will sum more effectively. Thus, it is essentially impossible to determine exactly which structures of the brain are responsible for the recorded signal on the scalp (although modern computational methods can make reasonable guesses and approximations [29]). Despite this drawback, EEG allows for recording of brain activity with very high temporal resolution, on the order of milliseconds. This is essential for capturing rapid dynamics of brain activity including neural oscillations, or rhythmic neural firing at different frequencies that is critical for information processing [30]. This is also particularly useful for feedback-related applications, as interventions can be applied much more quickly.

EEG is also very useful for applied educational work because behavioral output from the learner is not required to obtain a signal – the brain will respond to stimuli whether or not a physical response is made. Decades of cognitive and neuroscientific research has demonstrated that EEG does in fact provide informative signal that can be leveraged. For example, induced power in the alpha rhythm of the EEG – a neural oscillation in the 8-12 Hz range – is related to attention and visual processing of information [31, 32]. Additionally, the phase of the ongoing alpha rhythm is predictive of successful processing of visual stimuli [33, 34]. Thus, by measuring ongoing alpha power, we can have some understanding of a learner's attentional state, and immediately modulate the virtual environment if attention is waning. Additionally, we can track the phase of the alpha oscillation to time the presentation of information and optimize its encoding.

EEG is not only useful for measuring broad state changes and low level perceptual processing, but is also effective at measuring higher level cognitive and mental processes. Predictive and reliable EEG measures of successful episodic memory encoding [35, 36], emotional operations [37], processing of semantics and meaning [38], and second-language learning [39] have been identified. Thus, EEG is not simply another method of measuring level of arousal; it potentially allows teachers to identify mental states and cognitive processes in real-time, making it an ideal tool for an educational feedback system.

4. PRELIMINARY DATA

We have described an idealized closed loop educational feedback system combining VR and EEG. The actual utility of such a system must be validated empirically. Additionally, combining EEG and VR presents multiple technical hurdles; EEG measurements must be sent in real-time to the VR environment controller, and EEG data artifacts produced by head movements and eyeblinks must be dealt with. Here, we present preliminary data from 12 participants during an experiment designed as a first pass to combine EEG and VR.

4.1 Experimental Design

The procedure was a visual working memory task developed with the Unity3D game engine. The task was completed by all participants both in VR and on in front of a computer screen with

a mouse. In a fairly simple virtual space, floating stimuli (truncated icosahedrons) appeared in random locations from a pre-defined set of possible locations. After appearing, the stimuli would change to random colors for 1 sec, and then turn back to gray for a delay of 1.1 sec. Following the delay, the floor of the virtual environment would change to a color that one of the stimuli had previously been. The participants then had to interact with a stimulus to indicate which of the stimuli presented had been that color. If correct, the stimulus exploded and the next trial began; if incorrect, the stimulus shrunk to nothing and the next trial began.

The task was adaptive, such that the number of stimuli presented was dependent on the participant's performance. At the start of the experiment, 3 stimuli were presented. If the participant got a trial correct, a counter would move up 1; if incorrect, down 1. If the counter reached +3 or -3, the number of stimuli would increase or decrease, respectively. Each participant completed 20 minutes of trials in the VR condition and 20 minutes in the non-VR condition.

In the non-VR condition, participants were seated in front of a computer monitor and interacted with stimuli by clicking a mouse. In the VR condition, participants wore an HTC Vive VR headset and interacted with stimuli using a handheld motion controller. The Vive uses room-scale tracking, so participants could walk up to stimuli and interact with them. In both conditions, participants wore an InteraXon Muse EEG headband during the task to record EEG data. The Muse is a consumer headset with 4 channels (2 frontal, 2 temporal), a 200 Hz sampling rate, and wireless capabilities. Although a consumer headset may not provide as clean of a signal as a medical grade EEG system, other work has shown that mental states can still be classified with the Muse [40], and demonstrating success with a consumer-grade headset provides ecological validity, as real-life educational applications will likely use consumer products.

Behaviorally, participants did not perform any better or worse in VR compared to non-VR. Previous studies have found benefits when performing tasks in VR, although these benefits are likely somewhat task-dependent, and simpler, less complex tasks may not lead to large improvements [41]. Thus, the simplicity of our working memory task may have reduced the benefits of VR. However, our primary goal was not to show large benefits of VR usage – this experiment represents a first step towards combining VR and EEG for neurofeedback work, and using a simple experiment with a learning component before moving to a more complex learning environment may allow us to identify informative EEG features.

4.2 EEG Data

Our goal with the EEG data was to identify neural features that differentiated trials in which participants made a correct response from trials in which they didn't. Importantly, we examined data during the encoding period of the task – while participants were viewing the colored stimuli prior to the delay period. Theoretically, this is when subjects were encoding information to be retained in memory. If neural signals already differentiate accurate and inaccurate choices at this time, then interventions could be applied in real-time to lead subjects to successful learning prior to a response even being made.

4.2.1 Event Related Potentials

We first examined event-related potentials (ERPs) in response to the onset of the stimuli to be remembered [42]. ERPs are

measured by baseline correcting the EEG data for each individual trial, and then averaging the trials together to create an average response related to the stimulus. We used a baseline period of 100 ms prior to the onset of the stimuli, and filtered the data with a 16 Hz low-pass Butterworth filter to reduce noise. Plotted in Figures 1 & 2 are the grand average ERPs (averaged across subjects) of correct vs. incorrect responses in the VR and non-VR conditions for the temporal channels. Frontal channels were not plotted, as the data were noisy and inconclusive.

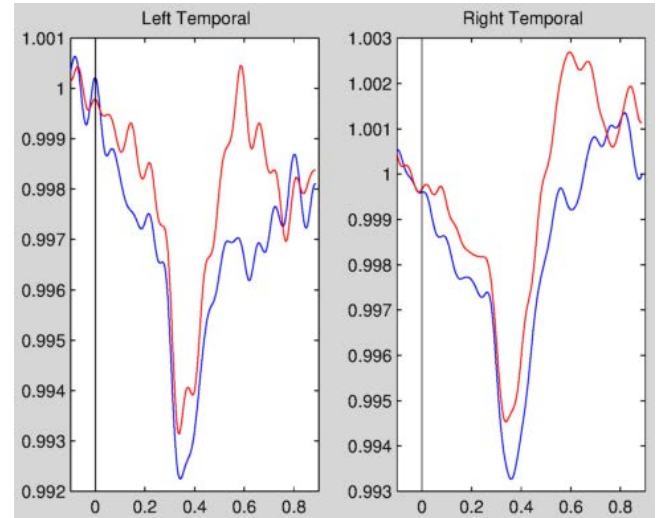


Figure 1. ERPs from the non-VR condition during the encoding period. Amplitude change is plotted on the Y axis, time (including baseline) is plotted in sec on the X axis. Stimulus onset is at time 0. Blue lines are correct, red lines are incorrect.

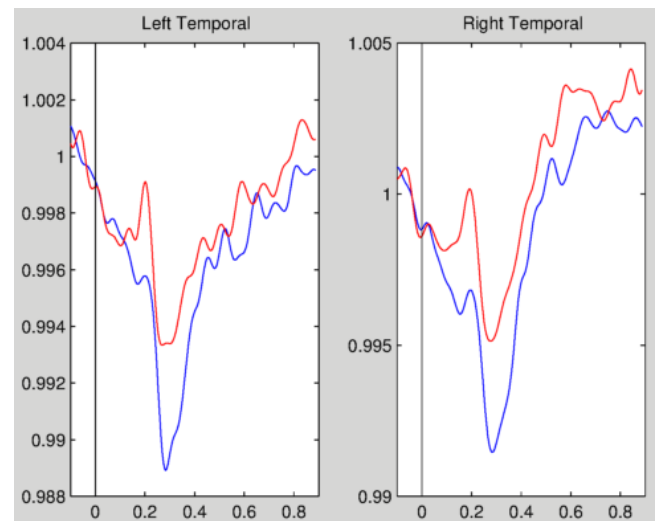


Figure 2. ERPs from the VR condition during the encoding period. Amplitude change is plotted in sec on the Y axis, time (including baseline) is plotted on the X axis. Stimulus onset is at time 0. Blue lines are correct, red lines are incorrect.

Looking at the ERPs, it is clear that there are differences between correct and incorrect responses during the time of encoding. Interestingly, the large negative-going component 300 ms after the onset of the stimulus is earlier in the VR condition, and the

differences between correct and incorrect are larger. A t-test on the mean amplitudes between 250-400 ms across correct and incorrect was trending toward significance ($p = 0.061$) in the right temporal channel. It is unclear why the magnitude of this effect would be larger in VR – while it is tempting to view it as a neural correlate of immersion or spatial awareness in VR, caution must be taken, as movement and eyeblink artifacts could contribute.

4.2.2 Time-Frequency Analysis

To examine stimulus related oscillatory activity, we performed time-frequency analysis [43]. We computed the Fast Fourier Transform (FFT) of 250 ms windows of the encoding period that were tapered with a Hanning function. Each window had an 87% overlap with the previous window. We also computed the FFT of a 250 ms pre-stimulus baseline for each trial. After averaging spectral data from each trial, we performed baseline correction by Decibel conversion ($10 * \log(\text{activity} / \text{baseline})$). Thus, the figures plotted display changes in power relative to baseline across the encoding time period for the temporal channels.

Examining figures 3 & 4, we can see similarities between the VR and non-VR conditions. First, there was an early decrease in alpha (12 Hz) power from approximately 0-300 ms, followed by a later increase in alpha power. Alpha effects were significant in the early time window in the left channel ($p = 0.0437$) and trending toward significance in the late time window ($p = 0.0543$) only in the VR condition. Additionally, in both VR and non-VR, there was a decrease in high beta (30-40 Hz) power from approximately 0-300 ms in correct responses relative to incorrect responses. Average power was computed at approximately 28-33 Hz and 38-43 Hz from roughly 0-270 ms in both channels, and these power values were submitted to t-tests. The lower range (28-33 Hz) effect was significant for non-VR ($p = 0.025$) and VR ($p = 0.044$) in the left channel only, while the higher range (38-43 Hz) effect was significant only for VR in the right channel ($p = 0.0428$).

5. CONCLUSION

Here we have given a theoretical overview of a closed-loop educational feedback system in which neurobiological signals from the learner shape the virtual learning environment to optimize learning. We also presented preliminary experimental data that suggests such a predictive model is feasible, as we were able to identify features in the EEG data that differentiated correct and incorrect responses at the time of encoding. Future efforts will attempt to use these features for machine learning classification of correct vs. incorrect responses, with the eventual goal of having a model that can accurately perform this discrimination in real-time and intervene to improve performance. Statistically, time-frequency responses were more robust, particularly in the VR condition, suggesting these features may be more useful.

Important technical hurdles must still be overcome – the main being removing artifacts from the EEG data. Eyeblinks and head movements can create large fluctuations that mask the informative signal. This problem is exacerbated when using a wireless commercial headset, as the occasional signal dropout and dropping of samples leads to even more noise. The Muse presented an additional problem with the frontal channels – the proximity of these channels to the reference, and to eye muscles, resulted in highly noisy data that was mostly uninformative. Additionally, with only a few electrodes, conventional methods for artifact removal such as ICA [44] are mostly unsuccessful. More advanced signal processing methods, such as adaptive filtering [45], will be necessary to achieve high classification accuracy, and will be critical for real-time implementation.

An important question is whether an EEG-VR intervention system would be the most optimal system for enhancing learning. Theoretically, interventions could occur before the participant even makes an error, and if the participant is unaware of their error, they may not learn as well. In other words, errors can be informative, and intervening before errors may not be the best thing to do. We propose that there are multiple reasons why an individual might make an error, including incorrect knowledge about the problem, and lack of attention or encoding. If the individual has incorrect knowledge, post-error learning might be more useful; however, if the error is simply due to lack of attention, a pre-error intervention might be more useful. An EEG-VR feedback system would allow for both types of interventions.

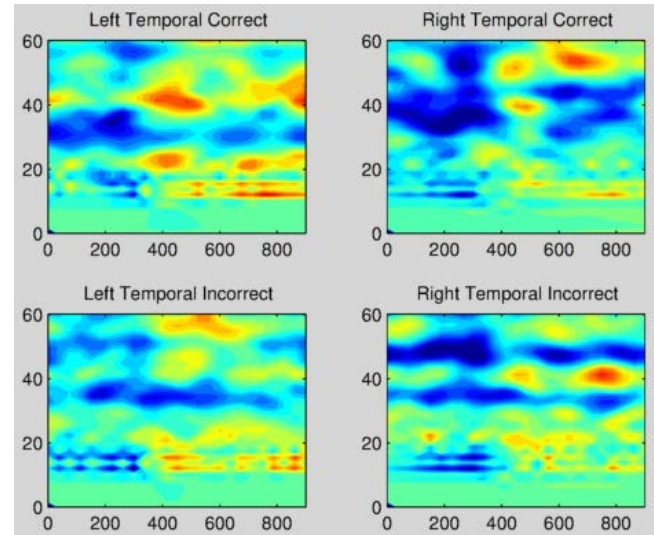


Figure 3. Time-frequency plots of the encoding period of the non-VR condition. Frequency (Hz) is plotted on the Y axis, time from onset of stimulus (ms) on the X axis, and color represents relative power. Baselines not plotted.

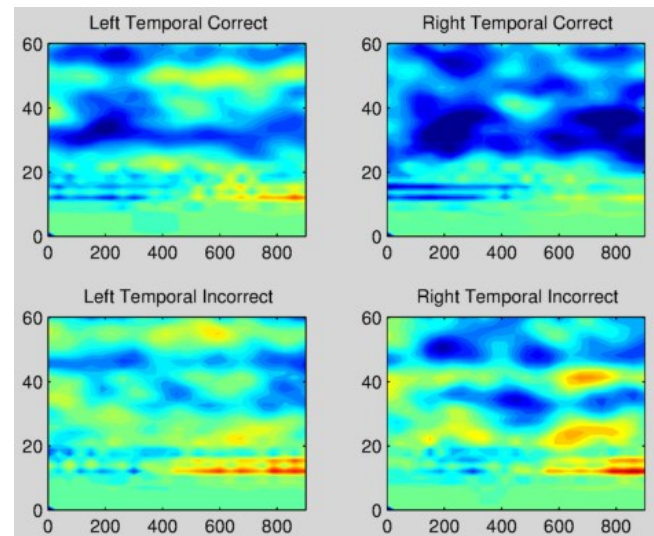


Figure 4. Time-frequency plots of the encoding period of the VR condition. Frequency (Hz) is plotted on the Y axis, time from onset of stimulus (ms) on the X axis, and color represents relative power. Baselines not plotted.

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